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ORIGINAL PAPER

A crucial remark on the importance of considering transient configurations on the optimal coordination of directional overcurrent relays in power systems with high penetration of inverter-based generation resources

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Abstract
Since 1988, several papers have been written on the topic of optimal coordination of directional overcurrent protections (DOCP). Nevertheless, a fundamental deficiency can be highlighted in the vast majority of these contributions, and in particular, in the optimization formulations used to solve the problem: the constraints associated with the selectivity criteria do not consider the dynamic changes in the network topology that appear during the fault clearing process. In this paper, we analyze the severe negative technical consequences of discarding this important aspect in the formulation of the optimization problem in the context of smart grid applications based on the integration of inverter-based generation resources in interconnected power systems. It is shown that the above-mentioned simplification leads to an important loss of selectivity and, therefore, to a miscoordinated relay operation. Three illustrative examples are presented and discussed, showing that the omission of the correspondent constraints jeopardizes the security and resiliency of the system in real-world conditions. Protection system selectivity is crucial to providing adequate reliability in the case of meshed networks with high penetration of distributed energy resources.

Keywords Performance indexes · Optimization of coordination of protections · Directional overcurrent relays · Dynamic configuration states

List of symbols

Superscripts

$+$	Denotes a simulated index
$*$	Denotes an optimum variable
$\dagger$	Denotes a corrected time due to dynamic states

Data sets & Matrices

$\mathbf{B}$	Coordination matrix
$\mathbf{D_{min}}$	Set of minimum dial times
$\mathbf{D_{max}}$	Set of maximum dial times
$\mathbf{I}$	Set of fault current magnitudes
$\mathbf{D}$	Set of time dial settings
$\mathbf{P}$	Set of pick-up current settings
$\mathbf{S}$	Set of separation times
$\mathbf{S_M}$	Set of safety margins

Variables

α	Relay curve parameter
$\overline{\nu}$	Average speed index
I	Fault current magnitude
N	Number of analyzed pairs
N_{nosen}	Pairs with no-sensitive relay pairs
N_{sel}	Selective relay pairs
N_{nosel}	Selective relay pairs
N_F	Pairs with computable separation time
n_l	Number of transmission lines
n_{pr}	Number of total primary backup relay pairs

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n_f	Number of faults locations per line
n_{fs}	Number of simulated faults per line
n_r	Number of relays
D_r	Time dial setting of relay r
P_r	Pick-up current setting of relay r
S	Separation time
S_M	Safety margin
D_{min}	Minimum dial time
D_{max}	Maximum dial time
T	Operation time

1 Introduction

Under the energy transition paradigm, current and future power systems are compelled to incorporate huge amounts of inverter-based resources (IBR). Many renewable-based systems must be interfaced with AC networks with power electronic converters with limited short-circuit contribution capability. To deal with low short-circuit levels, new network interconnections will appear, and several devices, such as FACTS, are going to be installed in so-called weak grids in order to preserve system protection and control.

Widespread installation of IBR is focused on existing subtransmission and distribution systems, traditionally operated as passive grids. As these grids have to evacuate large amounts of energy, reliability standards also impose the need to interconnect existing and future networks. In the context of developing and adapting protective systems for multiple source networks, the use of directional overcurrent protections (DOCP) has acquired renewed interest.

Since 1988 [1], important effort has been devoted to the topic of optimal coordination of DOCP, and several technical papers have been written. More recently, this topic has been regarded as part of the smart grid paradigm [2]. Adequate speed and selectivity of protective devices are crucial to incorporating large-scale power generation systems linked to weak networks [3–6]. This kind of protection has been extensively used for decades and offers important advantages as well as some disadvantages. The DOCP remains the leading protection scheme, especially for protection against line-to-ground faults [7]. The DOCP is very economical in comparison with other protection schemes, and, very importantly, it does not depend on communication links to clear the faults.

Existing approaches for the coordination of DOCP may be basically classified into two categories. On the one hand, there are linear models based on the selection of time dial or multiplier settings (TDS) that assume that the pick-up current multiplier settings have been previously calculated and known. On the other hand, there are complete nonlinear models where both the time dial and pick-up settings are regarded as unknown variables and calculated simultaneously. In general, the main objective is to minimize the overall operation time, subject to selectivity and sensitivity constraints.

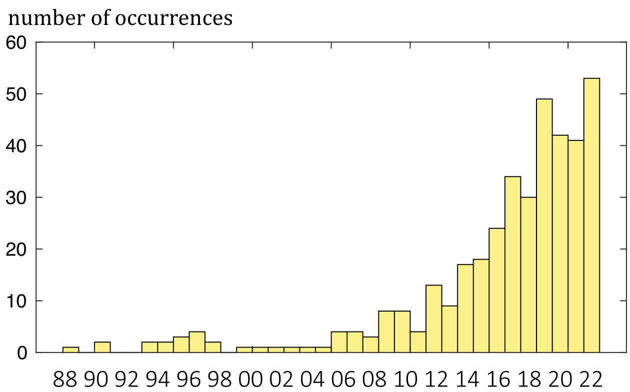
However, the optimal coordination of DOCP is still difficult and challenging in real-world applications since most existing models are based on simplistic assumptions. An important feature that can be highlighted in the vast majority of existing DOCP optimization formulations, in this case due to their absence, is the omission of dynamic state constraints associated with transient changes in the network topology during the clearing interval [8]. This condition produces a current redistribution that affects operation times and may lead to miscoordination if relays are configured with TDS obtained by simplistic optimization models that disregard the effect of dynamic states [9]. The dynamic states under consideration should not be confused with operational changes in network topology (reconfiguration) under steady-state conditions [10, 11].

As a matter of fact, most of the existing contributions on this topic assume that primary relays operate at the same time when a fault occurs in their protection zone. This assumption is not realistic since primary relays at both line ends of a transmission line always operate at different times, and consequently, the current abruptly changes in the slower relay and their backups in adjacent lines. As a result, the calculation of the separation times between backup and primary operation times is inaccurate, and comparisons with prescribed coordination time intervals are clearly biased.

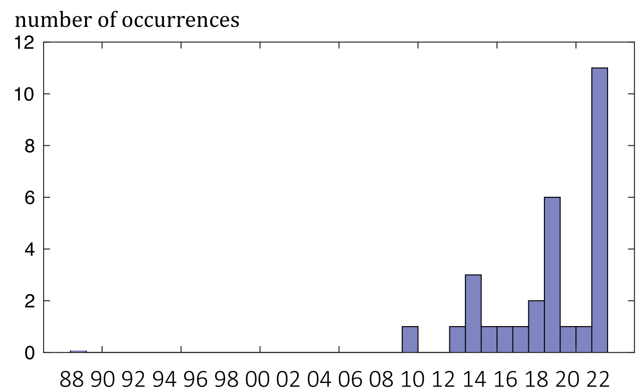
This situation is particularly significant in transmission systems with large penetration of IBR. Inverters often have limited short-circuit current capability compared to traditional generators. This means the fault current might not be much higher than the nominal current (e.g., 1.1 to 1.5 times) and strongly dependent on IBR control system interactions [12]. Two implications must be considered in the coordination of DOCP with IBR: 1) The pick-up setting might need to be lowered compared to what would be typical for a conventional generator, as the fault current will be lower. 2) The TDS may need to be finely adjusted to account for the lower fault current contribution. The fine tune of the TDS considering dynamic state changes is a challenging task that must resort to optimization to ensure selectivity.

To our knowledge, despite the fact that there are in the literature some papers on coordination of DOCP that consider dynamic network configurations [8, 13–17], the effect of not considering this subset of constraints has not been evaluated and analyzed, and there are few contributions considering systems dominated by inverter-based generation resources. A previous paper, [18] addressed the optimal coordination of DOCP but disregarding the effects of dynamic network reconfigurations.

To fill the research gap, in this paper, the technical consequences of omitting dynamic network configurations in the quality of the solution of the problem of coordinating DOCP



(a) Contributions on optimal coordination of DOCP



(b) Contributions on DOCP in the context of energy transition electrical grids

Fig. 1 Comprehensive review of contributions on optimal coordination of DOCP

Table 1 Relevant contributions on optimal coordination of DOCP

	year	Reference	Model type	Solver implemented	Consider dynamic states?	Consider IBR systems?
1	1988	Urdaneta et al [1]	LP	Simplex	no	no
2	1996	Chattopadhyay et al [19]	LP	Simplex	no	no
3	1997	Urdaneta et al [8]	LP	Simplex	yes	no
4	2000	So et al [13]	MHP	Evolutionary	yes	no
5	2003	Abyaneh et al [20]	LP	Simplex	no	no
6	2006	Zeineldin et al [21]	MHP	Particle Swarm	no	no
7	2006	Birla et al [22]	NLP	Sequential Quadratic Programming	no	no
8	2007	Mansour et al [14]	MHP	Particle Swarm	yes	no
9	2010	Thangaraj et al [23]	MHP	Evolutionary	no	no
10	2011	Badekar et al [24]	MHP	Genetic Algorithm	no	no
11	2012	Amraee et al [25]	MHP	Seeker	no	no
12	2013	Singh et al [26]	MHP	Teaching Learning	no	no
13	2015	Saleh et al [18]	NLP	Sequential Quadratic Programming	no	yes
14	2015	Albasri et al [27]	MHP	Biogeo	no	no
	2007	Birla et al [28]	NLP	Sequential Quadratic Programming	yes	no
	2016	Singh et al [15]	MHP	Differential search	yes	no
	2018	Nunez et al [29]	LP	Interior Point	yes	no
	2019	Mohammadzadeh et al [30]	LP	Interior Point	yes	no
	2020	Sorrentino et al [16]	LP	Interior Point	yes	no
	2020	Mahboubkhah et al [17]	LP	Interior Point	yes	no
	2024	Sorrentino et al [31]	LP	Interior Point	yes	no
	2024	De Oliveira et al [32]	LP	Monte Carlo Interior Point	yes	no
	2024	This proposal	LP	Interior Point	yes	yes

in interconnected power systems with IBR are analyzed. To do so, coordination solutions provided by optimization models, with [8, 16] and without [1] the consideration of dynamic states, are compared by performing detailed computer simulations to assess the quality of coordination solutions from selectivity and speed standpoints [9].

The contribution of this paper is summarized as follows: there is no precedent in the literature on the analysis of optimal coordination of DOCP with IBR sources considering dynamic network configurations during the fault clearance process.

The paper is organized in five sections. Background is presented in Sect. 2. The fundamentals of optimal coordination and simulation of DOCP systems and their application to systems with IBR are discussed in Sect. 3. Results and analysis are addressed in Sect. 4. Conclusion is presented in Sect. 5.

2 Background and related work

The topic of optimal coordination of DOCP was initiated in 1988 and, according to Web of Science records, has gained renewed interest since 2010, as shown in Fig. 1a. Comprehensive reviews on this topic can be found in [33–36]. The surge of publications on optimal coordination of DOCP coincides with the emergence of contributions where the use of DOCP systems is linked to up-to-date topics such as smart grids, renewable generation, active networks, and energy transition systems, as seen in Fig. 1b.

Table 1 lists the first 14 most influential publications (with higher number of citations) on the topic of optimal coordination of DOCP, according to the Web of Science [1, 8, 13, 14, 18–27]. Characteristics such as year, optimization model, and whether the proposal includes or does not include dynamic configurations are indicated. The problem has been addressed using different approaches, such as Linear Programming (LP) methods as Simplex and Interior point (IP), Nonlinear Programming (NLP), as Sequential Quadratic Programming (SQP), and Meta-Heuristics Programming (MHP) methods as Particle Swarm (PS), Biogeo (B), Evolutionary (E), Genetic Algorithm (GA), Seeker Algorithm (SA), Teaching Learning (TL), and Differential search (DS). It is important to mention that, beside contributions [8], [13], and [14], there are also only a few recent ones covering the topic of considering dynamic configuration states [15–17, 28, 31, 32]. Furthermore, some recent contributions have been proposed, taking into consideration the integration of inverter-based renewable resources [18], microgrids [2], and smart grids [3, 37].

However, most of the proposed models are based on the simplistic assumption that both relays operate simultaneously. According to [8], dynamic network configuration states should be categorized depending on the ability of primary and backup relays to detect faults during the first transient period. As seen in Table 1, this aspect is disregarded in the vast majority of existing contributions to this topic, either considering or not inverter-based generation resources.

3 Methodology

In this section, we analyze and compare two mathematical formulations for the statement of the optimization problem. In both models, the fault current contributions associated with inverter-based resources (IBR) are included. The first optimization model is the basic linear programming formulation presented in [1], while the second model comprises a more complete formulation that considers the dynamic changes in network topology during the fault clearing process proposed by [8] and refined by [16]. Finally, in order to evaluate the quality of coordination solutions given by each approach with IBR, a statistical tool based upon detailed, extensive simulations is used [9].

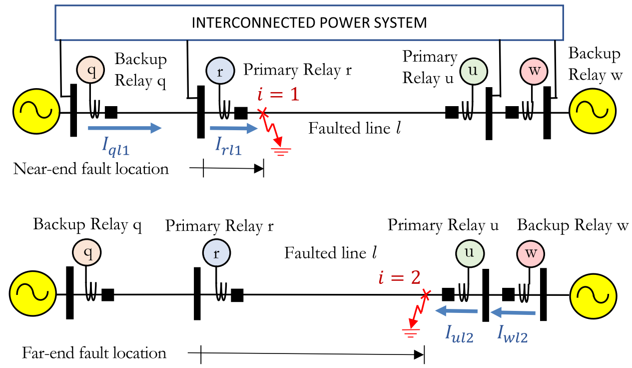


Fig. 2 Faulted line at two close-in locations

3.1 Basic DOCP coordination model

Consider an interconnected power system with $l = 1, \ldots, n_l$ transmission lines and n_r=$2n_l$ relays (two per line). The power system model comprises transmission line impedances and susceptances, network topology, power load, synchronous generation, inverter-based resource data, and relaying data. Depending on system topology, we can identify n_{pr} backup–primary relay pairs to be coordinated. For the sake of simplicity, only directional phase relays (67P) exposed to solid three-phase faults are considered. Future research must be devoted to analyzing other DOCP types (e.g., 67N) exposed to high-impedance line-to-ground faults.

Let us formulate the optimization problem considering two close-in fault locations per line: near-end (i=1) for left-handed relays and far-end (i=2) for right-handed relays, as shown in Fig. 2. Dynamic changes in network topology during the fault clearing process are not considered in this simplified approach. In the following, bold entries are regarded as arrays and matrices.

Primary relays r and u must have at least one backup relay q and w in adjacent lines, as shown in Fig. 2. As the pick-up current settings of backup and primary relays P_q, P_w, P_r, and P_u are known parameters in amperes, linear coefficients β_{qli}, β_{wli}, β_{rli}, and β_{uli} can be determined by Eqs. (1-2) [9]:

$$\beta_{qli} = \frac{\alpha_1}{\left(\frac{I_{qli}}{P_q}\right)^{\alpha_2} - 1}, \quad \beta_{rli} = \frac{\alpha_1}{\left(\frac{I_{rli}}{P_r}\right)^{\alpha_2} - 1} \tag{1}$$

$$\beta_{wli} = \frac{\alpha_1}{\left(\frac{I_{wli}}{P_w}\right)^{\alpha_2} - 1}, \quad \beta_{uli} = \frac{\alpha_1}{\left(\frac{I_{uli}}{P_u}\right)^{\alpha_2} - 1} \tag{2}$$

$\forall q, r, w, u = 1, \ldots, n_r, \forall l = 1, \ldots, n_l, i = \{1, 2\}$

where the subscript l denotes a faulted transmission line and i denotes a fault location. Notice that relays $q\&w$ and $r\&u$ are operating as backup and primary relays for faults located at i in line l, respectively.

Parameters α_1 and α_2 define the type of time-inverse curve used by each relay. For each close-in relay fault location, a short-circuit study should be carried out in order to determine the current contributions of the primary and backup relays of interest. Thus, terms I_{rl1}, I_{ul1}, I_{ql2}, and I_{wli2} are the corresponding fault current magnitude contributions (in rms amperes) associated with primary and backup relays for two close-in fault locations per line ($i = 1$ for r and u relays, and $i = 2$, for q and w relays), respectively. Short-circuit studies must include proper calculation of fault current contributions from inverter-based resources. The semiconductor switching devices of IBRs are intolerant of overcurrent, requiring to limit the fault current contributions to $1.1-1.5$ of the inverter nominal current rating. IBRs do not produce the large fault currents typical of synchronous generators [12]. Thus, short-circuit calculations required to compute linear coefficients β_{qli}, β_{wli}, β_{rli}, and β_{uli} in Eqs.(1-2) must include these reduced fault current contributions.

For each relay pair $q-r$ and $u-w$ at line l and fault location i, separation times S_{qrli} and S_{uwli} (time difference between backup and primary relays in fault locations 1 and 2) must be calculated as a function of relay characteristics, short-circuit currents flowing through backup and primary relays, and corresponding time dial settings. Each calculated separation time must be greater than a given safety margin S_M (sometimes called coordination time interval, CTI) specified by the protection engineer:

$$S_{qrl1} = T_{ql1} - T_{rl1} = \beta_{ql1} D_q - \beta_{rl1} D_r \geq S_M \quad (3)$$

$$S_{uwl2} = T_{wl2} - T_{ul2} = \beta_{wl2} D_w - \beta_{ul2} D_u \geq S_M \quad (4)$$

$$\forall q, r, w, u = 1, \ldots, n_r \forall l = 1, \ldots, n_l$$

where T_{ql1}, T_{rl1}, T_{wl2}, and T_{ul2} are the operation times at relays q, r, w, and u, for close-in fault locations in line l, respectively. Operation times rely on linear dependence between β_{ql1}, β_{rl1}, β_{ul2}, β_{wl2}, and time dial settings D_q, D_r, D_u, D_w, respectively.

The objective of the optimization problem is to maximize the overall speed of the protective system. This attribute is achieved by determining the set $\mathbf{D}^*$ of n_r time dial settings such that it minimizes the sum of tripping times of all relays acting as primary protections (T^*). The objective is subject to selectivity inequality constraints defined in Eqs.(3-4) as well as inequality constraints associated with time dial bounds (D_{min} and D_{max}).

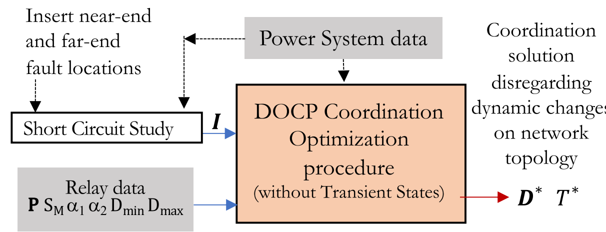


Fig. 3 Basic DOCP optimization process [1]

The model is scripted in matrix form in Eqs. (5-7) [1]:

$$\min_{\mathbf{D}^*} T^*$$

subject to: (5)

$$\mathbf{S} = \mathbf{B} \cdot \mathbf{D} \geq \mathbf{S_M} \quad (6)$$

$$\mathbf{D_{min}} \leq \mathbf{D} \leq \mathbf{D_{max}} \quad (7)$$

The optimal operation time T^* corresponds to the sum of n_r specific operation times of relays behaving as primary for close-in fault locations across n_l lines.

$$T^* = \sum_{l=1}^{n_l} \sum_{r=1}^{n_r} T_{rl1} + \sum_{l=1}^{n_l} \sum_{u=1}^{n_r} T_{ul2} \quad (8)$$

$$T^* = \sum_{l=1}^{n_l} \left[\sum_{r=1}^{n_r} \beta_{rl1} D_r + \sum_{u=1}^{n_r} \beta_{ul2} D_u \right] \quad (9)$$

Terms T_{rl1} and T_{ul2} are the operation times of relays r and u on the role of primary relays when a fault is located at locations 1 and 2 at line l. The settings of relays r and u are (D_r and D_u), time dials regarded as unknown variables, and (P_r and P_u), pick-up current settings regarded as known parameters. Please note that terms β_{rl1} and β_{ul2} are given by Eqs.(1 and 2) and may be calculated using other mathematical expressions for the inverse time curves.

Set $\mathbf{S}$ comprises n_{pr} separation times determined according to Eqs. (3-4) only for close-in relay fault locations in each line. Array $\mathbf{S_M}$ corresponds to the safety margin times previously established by the protection engineer. Each line of the coordination matrix $\mathbf{B}$ gathers linear elements (β_{ql1}, β_{wl2}, β_{rl1}, and β_{ul2}) shown in Eqs.(1-2) for backup and primary relays to be coordinated. Each element of the coordination matrix $\mathbf{B}$ should be determined considering fault contributions (magnitude and angle) in primary and backup relays. If a backup relay is unable to detect the fault due to loss of sensitivity or fault currents angles out of the detection zone, the corresponding relay pair row must be removed from the coordination matrix $\mathbf{B}$. Finally, each element of the coordination solution $\mathbf{D}^*$ must be enclosed between the lower (D_{min}) and upper limits (D_{max}). The resulting optimization problem has n_r unknowns and up to $n_{pr} + 2n_r$ inequality constraints.

Figure 3 shows a block diagram of the simplified optimization process. The gray box indicates the database elements required to carry out the optimization procedure. A short-circuit study is performed to get current contributions associated with primary and backup relays (**I**) for close-in fault locations. Afterward, the model is solved and results are printed.

3.2 Dynamic states DOCP coordination model

The formulation of the optimization problem considering dynamic states was introduced in 1997 by [8] and solved using linear programming. The fault clearing process has two well-defined time intervals: static and transient. First, when a fault occurs at location i, as seen in Fig. 4a, one of both ends is disconnected at time $T_{uli,0}$. The first period is denoted with the subscript "0" and defined as the *static interval*. In this interval, faster relay u advances toward the trip condition, clearing its fault current contribution at time $T_{uli,0}$= $\beta_{uli_0} D_u$.

This operation time depends on the fault contribution $I_{uli,0}$ observed during the static interval $0 < t < T_{uli_0}$ when the original network topology remains unchanged. During the static interval, the slower primary relay r and its backup q will advance if fault current angles are within the detection zone, and fault current magnitudes $I_{rli,0}$ and $I_{qli,0}$ are higher than the prescribed pick-up current settings P_r and P_q, respectively. Short-circuit studies must include proper calculation of fault current contributions from inverter-based resources.

Once the breaker associated with the faster primary relay u clears its fault contribution $I_{uli,0}$ at T_{uli_0} as seen in Fig. 4b, the network topology is modified, and abrupt changes on current magnitudes $I_{rli,1}$ and $I_{qli,1}$ are observed. In this case, the reminder period is denoted with the subscript "1" and defined as the *transient interval*. The transient interval lasts until the primary relay r clears the fault at a new time $T^{\dagger}_{rli}$, or until the backup relay q clears the fault at a new time $T^{\dagger}_{qli}$ if the primary relay r did not operate. Resulting operation times $T^{\dagger}_{rli}$ and $T^{\dagger}_{qli}$ must incorporate the effects of current magnitude changes occurring in both intervals, as indicated in [8, 16].

Six different selectivity conditions can be identified due to the inclusion of dynamic states [9]:

Type 1: Both relays of faster pair w-u advance toward the trip condition during the static interval. The separation time S_{wuli} on relay pairs w and u is computed with Eq.(10) [9]:

$$S_{wuli} = T_{wli,0} - T_{uli,0} = \beta_{wli,0} D_w - \beta_{uli,0} D_u \tag{10}$$

where $\beta_{wli,0}$ and $\beta_{uli,0}$ must be determined with currents $I_{wli,0}$ and $I_{uli,0}$ during the static interval, for $i = 1, \ldots, n_f$ fault locations in each line l, respectively.

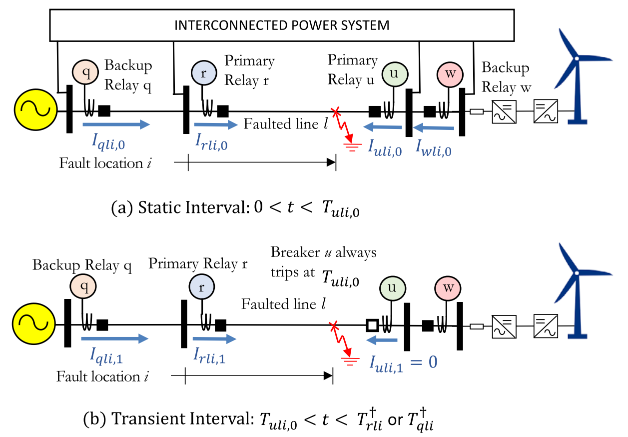


Fig. 4 Dynamic changes in network topology during the fault clearing process

Type 2: Both relays of slower pair q-r advance toward the trip condition during the static and transient intervals. The separation time S_{qrli} corresponds to the effective operation time $T^{\dagger}_{qli}$ minus the effective operation time $T^{\dagger}_{rli}$. The effective operation times take into account the transition between the static interval depicted in Fig. 4a and the transient interval shown in Fig. 4b. According to [9] $S_{qrli} = T^{\dagger}_{qli}$ - $T^{\dagger}_{rli}$ is given by:

$$S_{qrli} = \beta_{qli,1} D_q - \beta_{rli,1} D_r - \gamma_{uqrli} D_u \tag{11}$$

$$\gamma_{uqrli} = \beta_{uli,0} \left[\frac{\beta_{qli,1}}{\beta_{qli,0}} - \frac{\beta_{rli,1}}{\beta_{rli,0}} \right] \tag{12}$$

where entries $\beta_{qli,1}$, $\beta_{qli,0}$, $\beta_{rli,1}$, $\beta_{rli,0}$, and $\beta_{uli,0}$ must be determined with Eqs. 1 and 2 with currents $I_{qli,1}$, $I_{qli,0}$, $I_{rli,1}$, $I_{rli,0}$, and $I_{uli,0}$, respectively.

Type 3: Backup relay q does not advance due to loss of sensitivity and/or fault angle out of the detection zone, and it does not advance toward the trip condition during the static interval. In this case, the separation time S_{qrli} should be calculated using Eq.(11) but considering that the quotient $\beta_{qli,1}/\beta_{qli,0}$ is zero in the calculation of γ_{uqrli} (Eq. 12) [9].

Type 4: Primary relay r does not advance due to loss of sensitivity and/or fault angle out of the detection zone, and it does not advance toward the trip condition during the static interval. The separation time S_{qrli} should be calculated using Eq.(11) but considering that the quotient $\beta_{rli,1}/\beta_{rli,0}$ is zero in the calculation of γ_{uqrli} Eq.(12) [9].

Type 5: Both relays q and r do not advance due to loss of sensitivity and/or fault angle out of the detection zone, and they do not advance toward the trip condition during the static interval. The separation time S_{qrli} should be calculated using Eq.(11), but considering that γ_{uqrli} is zero [9].

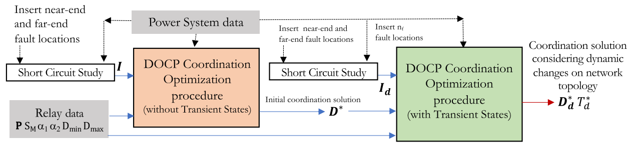


Fig. 5 DOCP optimization process considering dynamic changes on network topology

Type 6: Backup relays w or q do not advance due to loss of sensitivity and/or fault angle out of the detection zone in the transient period. No separation can be calculated [9].

The optimization model has the same structure as the simplified model stated in Eqs. (5-7), but with different objective and coordination matrices [16]:

$$\min_{\mathbf{D_d^*}} T_d^* \tag{13}$$

subject to:

$$\mathbf{S_d} = \mathbf{B_d} \cdot \mathbf{D_d} \geq \mathbf{S_M} \tag{14}$$

$$\mathbf{D_{min}} \leq \mathbf{D_d} \leq \mathbf{D_{max}} \tag{15}$$

The goal is to find out the dial settings $\mathbf{D_d^*}$ that minimize overall operation time T_d^* with transient configurations. The objective function corresponds to the sum of n_r specific operation times at the first interval of relays behaving as primary for close-in relay fault locations across n_l lines [16]:

$$T_d^* = \sum_{l=1}^{n_l} \sum_{r=1}^{n_r} T_{rl1,0} + \sum_{l=1}^{n_l} \sum_{u=1}^{n_r} T_{ul2,0} \tag{16}$$

where r and u are primary relays of line l

$$T_{rl1,0} = \beta_{rl1,0} D_r \tag{17}$$

$$T_{ul2,0} = \beta_{ul2,0} D_u \tag{18}$$

For each fault location $i = 1$ (near-end), 2 (far-end), $3, \ldots, n_f$ short-circuit study is carried out in order to determine the current contributions associated with primary and backup relays $\mathbf{I_d}$ during both static and transient intervals. The coordination matrix $\mathbf{B_d}$ is built using short-circuit currents. Each line of the coordination matrix $\mathbf{B_d}$ gathers the β and γ linear elements of the corresponding relay pair type (from 1 to 5) to be coordinated. Relay pairs where no separation time can be calculated must not be included in the coordination matrix $\mathbf{B_d}$ (Type 6). Arrays $\mathbf{S_d}$ and $\mathbf{S_M}$ comprise separation times and safety margin times, respectively. Finally, each element of the coordination solution $\mathbf{D_d^*}$ must be between the lower (D_{min}) and upper limits (D_{max}). The resulting optimization problem has n_r unknowns and up to $n_f \times (n_{pr} + 2n_r)$ inequality constraints. The inclusion of n_f=20–30 uniformly distributed fault locations per line in the model is enough to achieve a good selectivity level.

Figure 5 shows a block diagram of the optimization process considering transient states. The gray boxes indicate the database elements required to carry out the optimization procedure. It is important to note that an initial coordination solution, such as the one obtained in Sect. 3.1, is necessary to setup the model, since we do not know a priori the resulting tripping sequence of the improved optimization solution. If the time dial set $\mathbf{D_d^*}$ leads to a different tripping sequence from the initial solution $\mathbf{D}^*$, the set of n_f fault locations must be modified until they match.

For each set of fault locations, a short-circuit study is carried out to determine the current contributions associated with the primary and backup relays $\mathbf{I_d}$ during both static and transient intervals.

3.3 Evaluation of the quality of optimal coordination solutions [9]

The basic optimization model discussed in Sect. 3.1 aims to minimize overall operation time for complete selectivity. It is assumed that the worst cases (from a selectivity viewpoint) are covered by selecting close-in relay locations in each line. However, if the optimization model accounts transient states (as done in Sect. 3.2), actual separation times may be shorter and below the established safety margin, leading to miscoordination.

To investigate the consequences of the implementation of a given optimal coordination solution in real-world conditions, it is necessary to perform a statistical evaluation of the resulting speed, sensitivity, and selectivity indexes considering a representative number of simulated faults locations. The process for evaluation of the quality of a given coordination solution $\mathbf{D}^*$ (without transient states) or $\mathbf{D_d^*}$ (with transient states) is depicted in Fig. 6:

In [9], a comprehensive simulation methodology to assess performance indexes for selectivity, speed, and sensitivity is presented. A total of $i = 1, \ldots, n_{fs}$ uniformly distributed faults are simulated in each transmission line in order to determine the operation times as well as the separation times between backup and main relays. For each fault, short-circuit calculations are carried out, and the tripping sequence of each relay pair is identified and classified according to a number of possible outcomes from Types 1 to 6 (defined in Sect. 3.2). In this case, a separation time set at $\mathbf{S}^+$ is determined for any coordination solution with ($\mathbf{D} = \mathbf{D_d^*}$) or without transient states ($\mathbf{D} = \mathbf{D}^*$) with Eq. (19).

$$\mathbf{S}^+ = \mathbf{B}^+\mathbf{D} \tag{19}$$

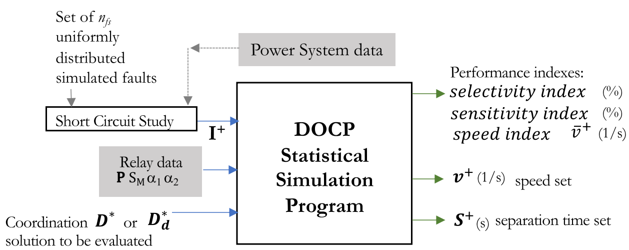


Fig. 6 Performance indexes evaluation/simulation [9]

where the coordination $\mathbf{B}^+$ used for simulation purposes has a dimension of $N_F \times n_r$. Term N_F is the number of relay pairs where a separation time can be calculated. In the remainder of the paper, the resulting separation times obtained from the comprehensive statistical analysis are denoted with the plus symbol ($^+$).

The selectivity index (%) is defined as the portion of set the $\mathbf{S}^+$, with length N_{sel}, whose elements (separation times) are greater than a prescribed safety margin ($S > S_M$) [9].

$$\text{selectivity index} = \frac{N_{sel}}{N_F} = 1 - \frac{N_{nosel}}{N_F} \tag{20}$$

For instance, a selectivity index of 90 % implies that 10 % of the occurrences (N_{nosel}) of calculated separation times (difference between backup and primary relay operation times) are lower to the prescribed safety margin S_M. Note that N_{sel}+N_{nosel}=N_F

The speed index in (1/s) is defined as the average speed $\overline{v}^+$ of n_r relays acting as primary protection when n_{fs} uniformly distributed faults are inserted in each line of the interconnected power system [9].

$$\text{speed index} = \overline{v}^+ = \frac{1}{2n_l n_{fs}} \sum_{l=1}^{n_l} \sum_{i=1}^{n_{fs}} \left[\frac{1}{T_{rli}^{\dagger}} + \frac{1}{T_{uli,0}} \right] \tag{21}$$

where main relays r and u are located at faulted line l. Terms $T_{rli}^{\dagger}$ and $T_{uli,0}$ are determined with Eqs. (22-24):

$$T_{uli,0} = \beta_{uli,0} D_u \text{ for Type 1 pairs} \tag{22}$$

$$T_{rli}^{\dagger} = \beta_{uli} D_u + \beta_{rl1,1} \left(D_r - \frac{\beta_{ul1,0}}{\beta_{rli,0}} D_u \right) \text{ for T2 \& 3} \tag{23}$$

$$T_{rli}^{\dagger} = \beta_{uli,0} D_u + \beta_{rli,1} D_r \text{ for Type 4 and 5} \tag{24}$$

Sets of simulated separation times $\mathbf{S}^+$ and speeds $\boldsymbol{v}^+$ can be graphically represented by means of histograms and other metrics.

Finally, the sensitivity index (%) is defined as the portion of the analyzed relay pairs able to perform the backup function without sensitivity problems [9]:

$$\text{sensitivity index} = 1 - \frac{N_{nosen}}{N} \tag{25}$$

where term $N = n_l n_{pr}$ is the total number of analyzed relay pairs and N_{nosen} accounts for no-sensitive backup relays during static and/or transient intervals. For instance, a sensitivity index of 90 % implies that 10 % of backup relays see a fault current (either on period 1 or 2) below the prescribed pick-up setting.

3.4 Impact of IBR fault currents on DOCP coordination

We aim to analyze the technical consequences of omitting the transient states in the DOCP optimization problem when IBRs are connected to the grid. As the IBR's fault contributions are similar to nominal loads, we investigate if it is worth including them or not in the optimization models. Thus, a comparison of the quality of coordination solutions provided by optimization models is carried out, with and without the consideration of transient states, and with and without the inclusion of IBR fault currents in the formulation of the problem. The quality of the obtained coordination solutions is exhaustively evaluated by means of the calculation of performance indexes for expected selectivity, sensitivity, and speed.

The analysis with and without the inclusion of IBR follows four steps:

Step 1: The basic optimization problem described in Sect. 3.1 is formulated as a linear programming (LP) problem disregarding dynamic changes in system topology during the fault clearing process with close-in relay fault locations in each line [1].

Step 2: Once the basic LP optimization problem is solved in step 1, we investigate the quality of the solution using a statistical test proposed by [9] and described in Sect. 3.3. Metrics for the whole protection system, i.e., sensitivity, selectivity, and speed (means and variances), are determined for several different fault locations, taking into account the effects of dynamic changes on system topology during the fault clearing process. In this step, we verify if time differences, or separation times, in all backup and primary relays are higher than a defined safety margin or allowed coordination time interval, irrespective of the fault location in the system. Some relevant miscoordination situations are analyzed in detail. We determine the selectivity degree and highlight the specific situations where the selectivity condition is not met.

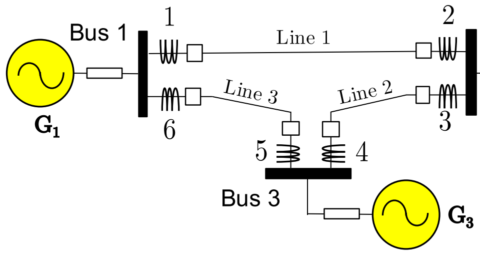


Fig. 7 Case study 1: 3-bus–3-line test system [1]

Step 3: As results in step 2 may show relevant miscoordination situations, we solve the LP optimization problem considering n_f uniformly distributed faults per line as well as the effects of dynamic changes on system topology during the fault clearing using the formulation provided by [16] described in Sect. 3.2.

Step 4: The quality of coordination solutions obtained in step 3 is also evaluated using the same comprehensive statistical analysis indicated in step 2.

4 Results and analysis

The proposed analysis is carried out in three case studies following the step-by-step analyses listed in Sect. 3.4. All optimizations and simulations were carried out on an Apple MacBook Pro with a 2.6 GHz Intel Core i7 with 16 MB of RAM.

4.1 Case study 1: 3-bus & 3-lines & 6 relays

This study case corresponds to the classical 69kV, 3-bus/3-lines/6-relays system presented in [1] and depicted in Fig. 7. This simple exercise allows us to show how the use of standard optimization that neglects transient states will lead to strong miscoordination. The solution is straightforward, and the interested reader can easily replicate the results using an Excel spreadsheet with a Simplex solver included in the following GitHub page:
https://github.com/pmdeoliveiradejesus/Optimal_DOCP.git. In this case, the pick-up settings are known parameters, and the coordination problem can be solved using any LP solution tool.

All system data indicated in [1] remain unchanged except line 2 that is specified as a short line of 6 km.

4.1.1 Step 1: Basic LP model solution

First, we apply the basic LP approach described in Sect. 3.1. This approach disregards the effects of dynamic changes on system topology and assumes that close-in relay fault locations are enough to satisfy the selectivity criteria. According to Eqs. (5-7), the structure of the basic optimization model is:

$$\min_{\mathbf{D}^*} T = \beta_{111} D_1 + \beta_{212} D_2 + \beta_{321} D_3 + \beta_{422} D_4 + \quad (26)$$
$$+\beta_{531} D_5 + \beta_{632} D_6$$

subject to :

$$\begin{bmatrix} -\beta_{111} & 0 & 0 & 0 & \beta_{511} & 0 \\ 0 & -\beta_{212} & 0 & \beta_{412} & 0 & 0 \\ \beta_{121} & 0 & -\beta_{321} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\beta_{422} & 0 & \beta_{622} \\ 0 & 0 & \beta_{331} & 0 & -\beta_{531} & 0 \\ 0 & \beta_{232} & 0 & 0 & 0 & -\beta_{632} \end{bmatrix} \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \\ D_5 \\ D_6 \end{bmatrix}$$
$$\geq \begin{bmatrix} S_M \\ S_M \\ S_M \\ S_M \\ S_M \\ S_M \end{bmatrix}$$
$$D_{min} \leq D_r \leq D_{max} \quad r = 1 \ldots 6$$

where $D_{min} = 0.1$, $D_{max} = 1.1$, $S_M = 200$ ms. Linear coefficients are: β_{111}=2.4199, β_{212}=1.5719, β_{321}=2.5921, β_{422}=2.6371, β_{531}=1.6616, β_{632}=2.0878, β_{511}=2.9393, β_{412}=2.9009, β_{121}=5.5708, β_{622}=4.6821, β_{331}=2.8243, β_{232}=2.4590.

The optimization model has six unknowns and $3 \times 6 = 18$ inequality constraints. Solving Eq. (26) is straightforward. The coordination solution is $\mathbf{D}^*$ = [0.1129; 0.2032; 0.1655; 0.1791; 0.1610; 0.1438]. The value of the objective function is T^* = 2.0613s (sum of 6 relay operation times). We would like to note that the reported solution does not change if other fault locations are included in the model.

4.1.2 Step 2: Basic LP solution evaluation/simulation

The quality of the coordination solution $\mathbf{D}^*$ obtained in step 1 with the basic optimization approach is evaluated by means of an exhaustive statistical simulation method, previously described in Sect. 3.2. This evaluation includes the effects of dynamic changes on system topology.

Performance indexes for sensitivity, speed, and selectivity are determined by inserting n_{fs} = 1000 uniformly distributed faults per line. Thus, 6000 faults are inserted to evaluate the response of the protection system. As the protection system has n_{pr} = 6 backup–primary relay pairs when one fault is inserted in each line, a total of 6000 relay pairs are suitable to be analyzed. The simulation time was 18.67s.

The resulting sensitivity index is 95.15 %. This good result is associated with the correct selection of pick-up current settings (known parameters and not subject to optimization in this paper). A higher sensitivity index reveals an adequate choice of the current pick-up settings for the original test case analyzed in [1]. In our view, selecting the settings to solve the sensitivity problem depends on how the operator dispatches the generation at the time. For the sake of shortness, this matter is outside the scope of this paper.

The resulting average speed index is $\overline{\nu}^{+}$ = 2.406 1/s, corresponding to an average operation time per relay of 428 ms (26 cycles) with a standard deviation of 70 ms, as seen in the histogram shown in Fig. 8a. The use of SI curves provokes extremely long operation times.

Statistical analysis shows that the number of elements of $\mathbf{S}^{+}$ is N_F=5988. This means that almost the entire set of available relay pairs to be evaluated is feasible. In other words, all backup relays are able to clear the fault under different circumstances. In 78.9 % of the cases, both relays of pairs (w-u and q-r) advance toward the trip condition during the static interval, and backups w and q are capable of clearing the fault (Types 1 and 2). In 21.1 % of the cases, partial selectivity is observed. This partial selectivity condition is achieved when some backup relays q do not advance during the static interval due to a lack of sensitivity and/or fault angles outside of the detection zone and only advance toward the trip condition during the transient interval, just before the fastest relay u clears its fault contribution (Type 3). Only 0.2 % (12 relay pairs of 6000) are unable to clear the fault due to the inability of backup relays to clear the fault due to lack of sensitivity and/or fault currents flowing in the opposite direction (Type 6).

The resulting selectivity index is 93.2 % showing that N_{sel}=5578 of the 5988 feasible backup–primary relay pairs fulfills the selectivity criteria. In this case, N_{nosel}=413 of 5988 relay pairs are no selective having a separation time Eq. (3) below the safety margin of 200 ms, as shown in the shaded area of the histogram depicted in Fig. 8b. Histograms only show occurrences between −0.5 and 2 s to facilitate the readability of graphs. Notice that the distribution of operation and separation times does not follow a known probability distribution function.

An example of loss of selectivity is observed at fault location i=864 when the calculated separation time of 180 ms is lower than the prescribed safety margin of 200 ms. This case corresponds to relay pair 6–4 (Type 2) when a fault occurs at 86.6 % of line 2, between buses 2 and 3, as seen in Fig. 10. The faster primary relay 3 operates at $T_{32,864}$ = 0.46s. When the circuit breaker associated with primary relay 3 (the faster one) clears its fault contribution, slower primary and backup relays 4 and 6 see an abrupt change in their fault current contribution from 3080 to 3310 A, and from 832 A to 1070 A. Relay 4 advances to the trip condition at $T^{\dagger}_{42,864}$ = 0.478s, and relay 6 advances to the trip condition at $T^{\dagger}_{42,864}$ = 0.658s, almost 180 ms later. Thus, the resulting separation time is $S_{643,864} = T^{\dagger}_{62,864}$-$T^{\dagger}_{42,864}$ = 180 ms. This time is clearly lower than the prescribed safety margin applied in the basic optimization problem (S_M = 200 ms) solved in Sect. 3.1. This result is important since it demonstrates that the basic optimization approach is unable to guarantee complete selectivity criteria under realistic conditions.

4.1.3 Step 3: Dynamic states DOCP coordination model

In order to improve the selectivity index observed in the previous step, the optimal DOCP coordination problem should incorporate the effects of dynamic changes on system topology during the fault clearing process. This aspect is crucial to determining an adequate solution that ensures the highest levels of selectivity and speed.

The case study is analyzed considering n_f=2, near- and far-end fault locations per line. According to Eqs.(13-15), and using the coordination solution obtained in step 1 to build the initial coordination matrix tripping sequence, the improved optimization model is written as follows:

$$\min_{\mathbf{D}^*} T_d = T_{111,0} + T_{321,0} + T_{531,0} + T_{212,0} + T_{422,0} + T_{632,0}$$
$$\text{subject to :} \tag{27}$$

$$\begin{bmatrix} -\gamma_{14211} & -\beta_{211,1} & 0 & \beta_{411,1} & 0 & 0 \\ -\beta_{111,0} & 0 & 0 & 0 & \beta_{511,0} & 0 \\ 0 & 0 & -\gamma_{36421} & -\beta_{421,1} & 0 & \beta_{521,1} \\ \beta_{121,0} & 0 & -\beta_{321,0} & 0 & 0 & 0 \\ 0 & \beta_{231,1} & 0 & 0 & -\gamma_{51431} & -\beta_{631,1} \\ 0 & 0 & \beta_{331,0} & 0 & -\beta_{531,0} & 0 \\ \beta_{112,1} & -\gamma_{25112} & 0 & 0 & -\beta_{512,1} & 0 \\ 0 & -\beta_{212,0} & 0 & \beta_{412,0} & 0 & 0 \\ 0 & 0 & -\gamma_{36422} & -\beta_{422,1} & 0 & \beta_{622,1} \\ \beta_{122,0} & 0 & -\beta_{322,0} & 0 & 0 & 0 \\ 0 & 0 & \beta_{332,1} & 0 & -\beta_{532,1} & -\gamma_{63532} \\ 0 & \beta_{232,0} & 0 & 0 & 0 & -\beta_{632,0} \end{bmatrix} \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \\ D_5 \\ D_6 \end{bmatrix}$$
$$\geq \mathbf{S_M} D_{min} \leq D_r \leq D_{max} \quad r = 1 \ldots 6$$

where $\beta_{422,0}$=0.637, D_{min} = 0.1, D_{max} = 1.1, S_M = 200ms. Numerical values of linear coefficients β can be extracted from the coordination matrix $\mathbf{B_d}$ Eq.(28).

2.76

$$\mathbf{B_d} = \begin{bmatrix} \text{[redacted]} & -2.246 & 0 & 6.041 & 0 & 0 \\ -2.420 & 0 & 0 & 0 & 2.939 & 0 \\ 0 & 0 & 0.558 & -2.793 & 0 & 4.579 \\ 5.571 & 0 & -2.592 & 0 & 0 & 0 \\ 0 & 4.870 & 0 & 0 & 1.412 & -3.978 \\ 0 & 0 & 2.824 & 0 & -1.662 & 0 \\ -4.746 & 1.339 & 0 & 0 & 5.912 & 0 \\ 0 & -1.572 & 0 & 2.901 & 0 & 0 \\ 0 & 0 & 0.331 & -2.565 & 0 & 4.006 \\ 7.292 & 0 & -2.824 & 0 & 0 & 0 \\ 0 & 0 & 6.675 & 0 & -2.602 & 1.848 \\ 0 & 2.459 & 0 & 0 & 0 & -2.088 \end{bmatrix} \tag{28}$$

The coordination matrix $\mathbf{B_d}$ has 6 columns and 12 rows. Even rows correspond to Type 1 relay pairs. Rows 3 and 9

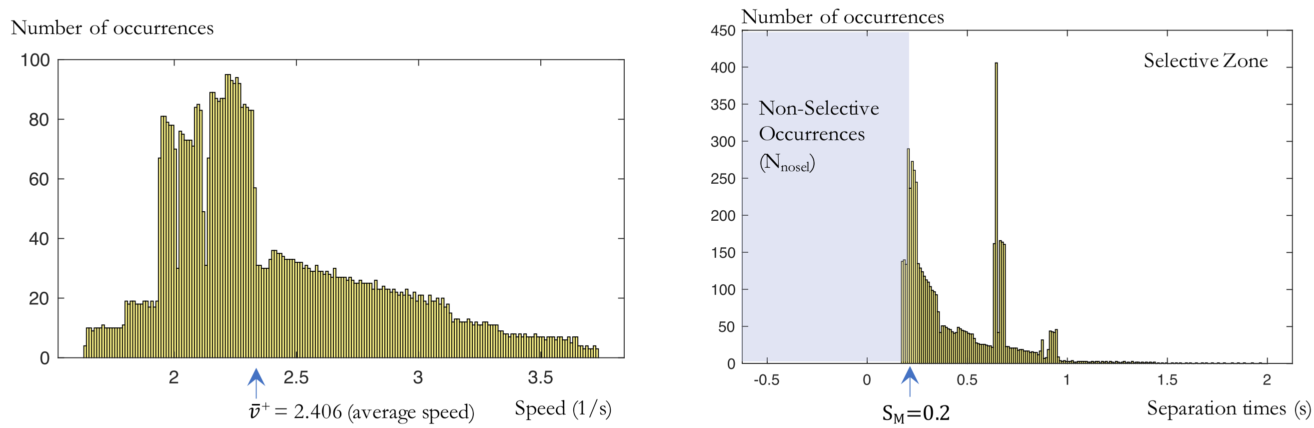


Fig. 8 Histograms for speed and selectivity for $\mathbf{D}^*$ settings (Case Study 1)

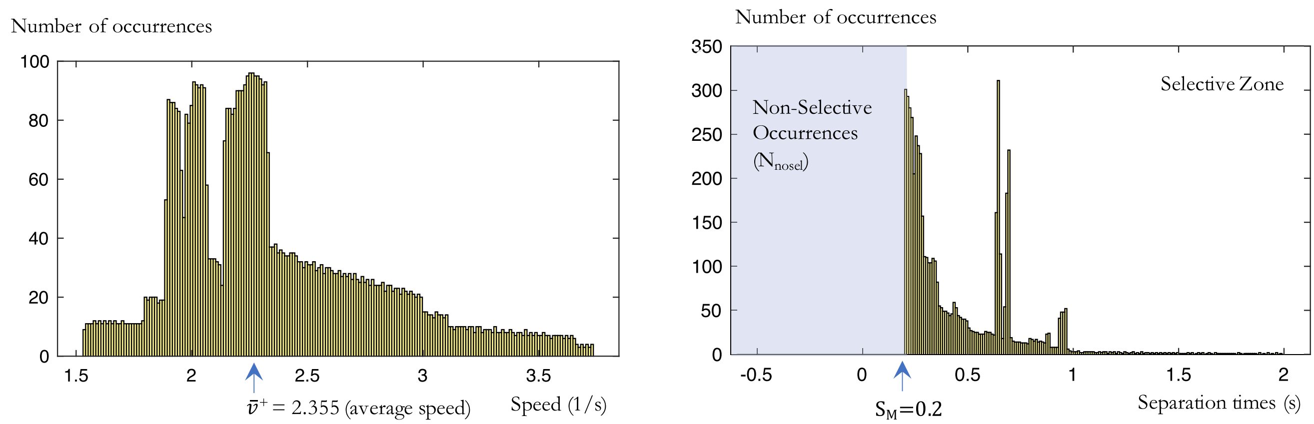


Fig. 9 Histograms for speed and selectivity for $\mathbf{D}_\mathbf{d}^*$ settings (Case Study 1)

are Type 2 relay pairs. Rows 1, 5, 7, and 11 are Type 3 relay pairs. The optimization model has six unknowns and 4 × 6 = 24 inequality constraints. The coordination solution to the complete optimization problem stated in Eq. (27) is $\mathbf{D}_\mathbf{d}^*$ = [0.1129; 0.2121; 0.1655; 0.1839; 0.1610; 0.1540]. The relay tripping sequence of the new solution $\mathbf{D}_\mathbf{d}^*$ is equivalent to the tripping sequence produced by the original coordination solution $\mathbf{D}^*$.

Notice that the time dial settings are slightly higher than the coordination solution obtained in step 2. The value of the objective function is $T_d^* = 2.6604$s. (6 primary relay operation times). The resulting higher time dials and operation times with respect to the basic coordination solution are compatible with the results obtained by [16]. The suitability of this solution in more realistic conditions is also evaluated in the next step.

4.1.4 Step 4: Dynamic states DOCP solution evaluation

The quality of the coordination solution $\mathbf{D}_\mathbf{d}^*$ obtained with the optimization approach that includes dynamic configuration states is evaluated with the same statistical method applied in step 2. As made in step 2, performance indexes are determined by inserting n_{fs}=1000 uniformly distributed faults per line. The CPU time required to evaluate 6000 faults and pair relay behaviors was 18.58s.

Statistical analysis shows that $\mathbf{S}^+$ has 5949 elements. We observe that in 79.4 % of the cases, both relays of pairs (w-u and q-r) advance toward the trip condition during the static interval (Types 1 and 2). In the reminder, in 20.6 % of the cases, there is partial selectivity (Type 3). Only 0.8 % (51 relay pairs out of 6000) are unable to clear the fault (Type 6). We observe a slight improvement (79.4 %) in the number of relay pairs that can effectively clear the faults with respect to the share obtained in step 2 (78.9 %).

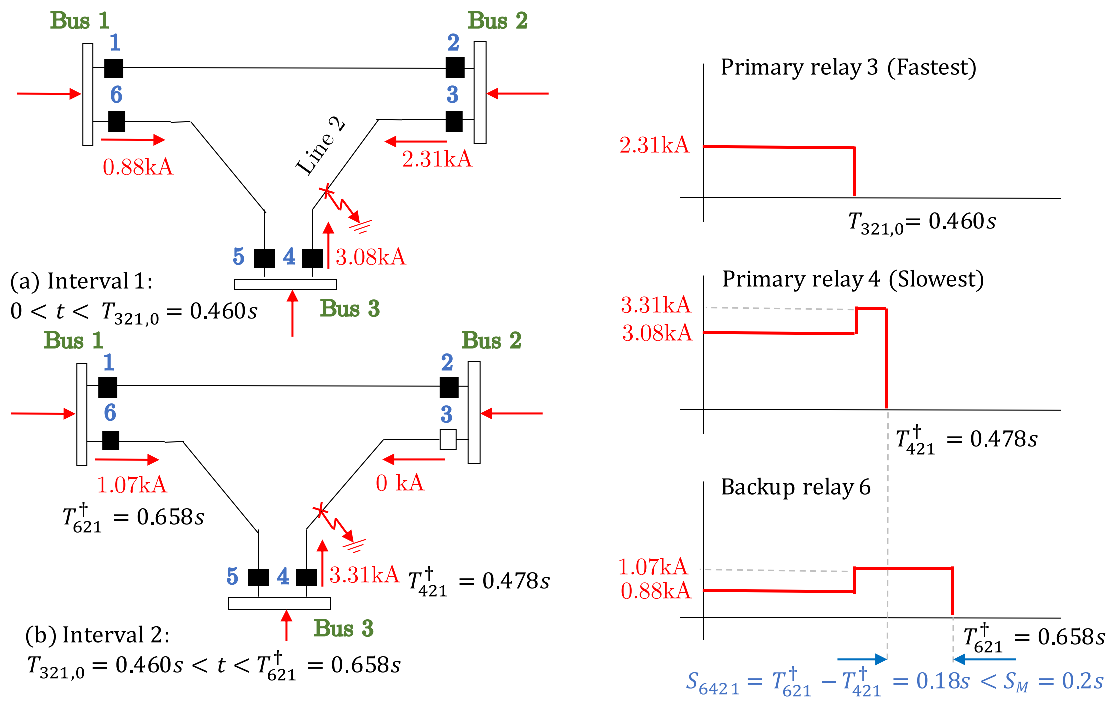


Fig. 10 Example of loss of selectivity in relay pair 6–4 with settings provided by the basic optimization formulation

As pick-up current settings do not change with the new time dial settings, the sensitivity index remains at 95.15 %. The average speed index is $\overline{v}^{\dagger}_{d} = 2.355$ 1/s, resulting in an average operation time per relay of 438 ms with a standard deviation of 75 ms, as seen in the histogram shown in Fig. 9a. These speeds are lower than the ones reported in step 2. The selectivity index shows that 100 % of the 5949 analyzed backup–primary relay pairs are suitable to clear fault currents. The coordination solution fulfills the selectivity criteria for all faults, irrespective of their location. In this case, all N_F=N_{sel}=5949 analyzed relay pairs produce a separation time higher than the safety margin of 200 ms, as shown in the histogram depicted in Fig. 9b. Thus, the inclusion of dynamic states enables complete selectivity, but with a reduction in the average speed of the DOCP. The inclusion of dynamic configurations and only two fault locations per line in the problem formulation is enough to guarantee complete selectivity in this illustrative example. However, this result is not general, as we can see in the second and third case study analyzed in this paper.

4.2 Case study 2: 4-bus & 5-lines & 10 relays

This example consists of a 115 kV 4-bus interconnected transmission system with two IBR. The one-line diagram is depicted in Fig. 11. Two synchronous generators of 100 and 75 MW are connected at buses 1 and 2 with a sub-transient reactance of 6.66 % and 13.33 %, respectively. All percentages and per unit values are expressed with 100 MVA and 115 kV bases. Wind and photovoltaic power stations of 25 and 12 MW are installed at buses 3 and 4, respectively. Short-circuit contributions of IBRs are 60 A and 120 A for solid three-phase faults at buses 3 and 4 (1.4 pu of their nominal currents), respectively. Four 100MVA 10/115kV step-up transformers are also installed in buses 1 to 4 with impedances of 3.5 %, 4.5 %, 4 %, and 6 %, respectively. Five transmission lines of 20, 25, 8, 20, and 35 km length have the same impedance, Z_L = 0.05+0.4i Ω per km. Line susceptances are neglected. Short-circuit calculations required to include IBR fault currents were performed using the Power Factory Digsilent tool (complete method) disregarding the effect of loads [38]. The

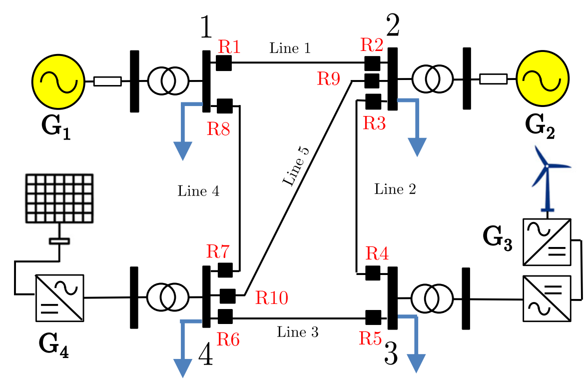


Fig. 11 Case study 2: 4-bus–5-line test system

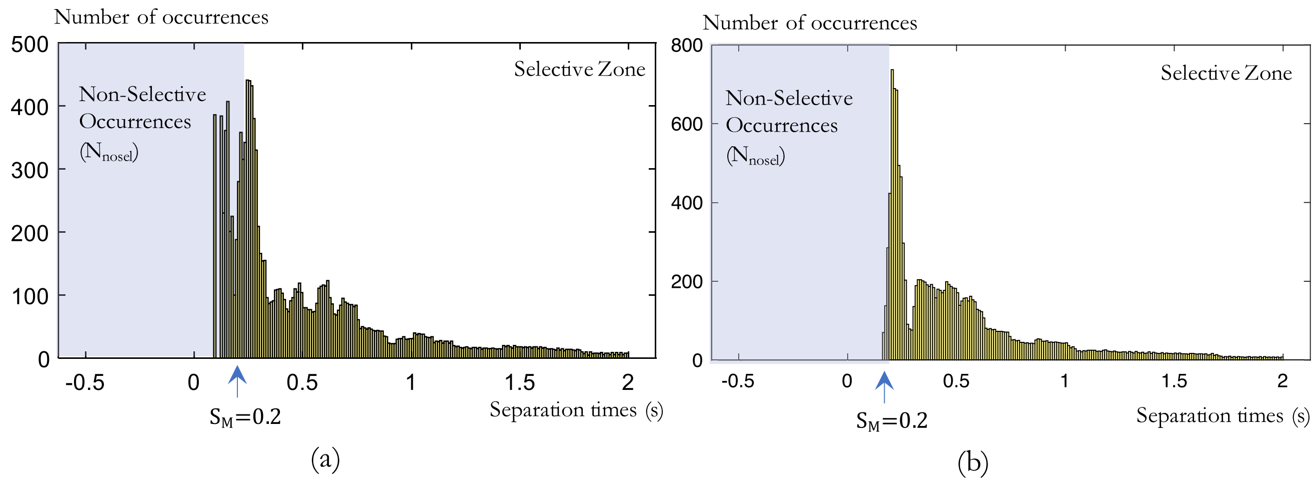


Fig. 12 Selectivity histograms for $\mathbf{D}^*$ and $\mathbf{D}_\mathbf{d}^*$ settings (Case Study 2)

optimization problem was solved using MATLAB's lingprog tool.

The protection of the transmission system comprises ten 67P-type directional overcurrent relays polarized at -45 deg $< \angle(VI^*) <$ +135 deg to detect faults within the line. The relay curve used is IEC 60,255 Extremely Inverse (EI). According to maximum load and generation conditions, considering n-1 contingencies, the pick-up current settings are set in kA as **P**=[0.2; 0.2; 0.25; 0.25; 0.2; 0.3; 0.3; 0.2; 0.2; 0.25] in order to assure adequate sensitivity. Time dial settings must be between 0.1 and 1.1. The number of relays n_r and relay pairs n_{pr} are 10 and 16, respectively. The safety margin adopted to formulate optimization problems is S_M = 200 ms.

The basic DOCP optimization formulation described in Sect. 3.2 (disregarding dynamic states) was applied to the 4-bus test system. The problem has 10 unknowns and 25 constraints. 11 backup–primary pairs were removed from the coordination matrix due to its inability to clear line faults. In these cases, separation times cannot be determined due to currents flowing in the opposite direction and/or loss of sensitivity at backup relays.

The optimal coordination solution (disregarding dynamic states) is $\mathbf{D}^*$ = [0.3394; 0.1748; 0.2324; 0.1000; 0.1590; 0.2822; 0.1000; 0.6673; 0.1421; 0.1000]. The value of the objective function is T^* = 3.3398s (10 primary relays operation times). The basic coordination solution was verified using the statistical simulator discussed in Sect. 3.3 considering n_{fs} = 1000 uniformly distributed faults. A total of 16,000 relay pairs behaviors' were evaluated, and the CPU time was 26.8s. The number of back–main relay pairs with calculated separation times is 14,783 (92.4 %).

Sensitivity, average speed $\overline{\nu}^+$, and selectivity indexes are 92.6 %, $\overline{\nu}^+$=12.728 1/s, and 73.7 %, respectively. The average operation time per relay (operating as the primary relay) is 102.7 ms, with a standard deviation of 43.4 ms. This means that, although under the basic optimization approach, the main relays are able to clear the faults faster (6 cycles), with those settings, a high loss of selectivity is obtained. A histogram for the selectivity results obtained with the basic formulation is shown in Fig. 12 (a). A severe miscoordination is observed; almost N_{nosel}=3890 of the 14,783 pairs that were analyzed do not fulfill the selectivity criteria. The worst case from a selectivity viewpoint corresponds to the 479th uniformly distributed faults, with a very low separation time of $S_{353,479}$ = 0.0782s. This separation time is observed on relay pair 3–5 (Type 2) when a fault occurs at 47.9 % of line 3 between buses 3–4 and it does not accomplish the predefined safety margin, $S_{353,479}$ = 0.0782s $< S_M$ = 0.2s. The result of 73.7 % for the selectivity index reveals poor performance, and it makes the current coordination solution not suitable. To improve this result, the optimization model that considers dynamic configurations described in Sect. 3.2 was applied to the 4-bus test system.

The detailed formulation (including dynamic changes in network topology) was scripted for n_f=21 fault locations per line uniformly distributed. The size of the coordination matrix is 420x10. The resulting coordination solution is $\mathbf{D}_\mathbf{d}^*$ = [0.4695; 0.1949; 0.4988; 0.1000; 0.2181; 0.3168; 0.1000; 0.7812; 0.1941; 0.1000] (the CPU time was 30.1s). The objective function is T_d^* = 0.6885s (10 primary relays operation times).

After running the simulator with n_f = 1000 to verify the quality of this solution, we can see that the coordination solution with 21 fault locations per line provides a significative improvement on the selectivity index from 73.7 % to 99.7 % [31]. In this case, simulation results show that the average operation time per relay increases from 102.7 to 128.4 ms. The standard deviation goes from 43.4 to 55.9 ms. In order to achieve the increased selectivity, the average speed of the DOCP fell from $\overline{\nu}^{+}$ = 12.728 1/s to $\overline{\nu}_{d}^{+}$ = 10.235 1/s.

A selectivity histogram for the results obtained with the improved formulation is shown in Fig. 12b. By comparing both histograms Fig. 12 (a) and (b), it is clear that the improved coordination solution avoids severe miscoordination with respect to solutions obtained with traditional DOCP formulations.

It is interesting to investigate what happens if we neglect IBR's fault contributions. The optimal coordination solution (disregarding dynamic states) and with no IBR installed in nodes 3 and 4 is $\mathbf{D}^{*}_{\mathbf{noIBR}}$ = [0.2500; 0.2500; 0.3769; 0.2847; 0.1700; 0.3347; 0.8873; 0.4124; 0.2888; 0.1000; 0.1070]. In this case, the simulation tool yields a selectivity index of 57.05 % and an average speed of 12.00 1/s. The detailed formulation (including dynamic changes in network topology) was scripted for n_f=21 fault locations uniformly distributed per line. The resulting coordination solution is $\mathbf{D}^{*}_{\mathbf{d-noIBR}}$ = [0.4608; 0.1797; 0.5129; 0.1000; 0.2217; 0.3029; 0.1000; 0.7917; 0.1966; 0.1000]. In this case, the simulation tool yields a selectivity of 99.60 % and an average speed of 9.69 1/s.

Notice that there are important differences between $\mathbf{D}^{*}$ and $\mathbf{D}^{*}_{\mathbf{noIBR}}$. Neglecting fault contributions of inverters is not a valid option using standard optimization since resulting selectivity is very low (57.05 %). When transient states are included, the resulting coordination solutions ($\mathbf{D}^{*}_{\mathbf{d}}$ and $\mathbf{D}^{*}_{\mathbf{d-noIBR}}$) are similar, but with a slight decrease in the selectivity index, from 99.70 % to 99.60 %. This result is meaningful, since IBR's fault contributions can be suppressed when advanced and detailed optimization are applied. The consideration of transient states is crucial to achieving high selectivity indexes.

4.3 Case study 3: IEEE 14-bus with 5 IBR

In 2014, [18] reported a DOCP coordination solution considering the well-known IEEE 14 test system including 5 IBR-based distributed generators connected to the 33 kV meshed sub-system as shown in Fig. 13. Optimal time dial settings reported for Case 1 correspond to a linear programming solution without transient states [1] with SI curves: $\mathbf{D}^{*}$ = [0.515; 0.309; 0.396; 0.600; 0.295; 0.442; 0.290; 0.366; 0.432; 0.316; 0.343; 0.282; 0.265; 0.400; 0.304; 0.610]. After running the simulator, the corresponding selectivity index is 79.43 % and the average speed index is 0.883 1/s. Strong miscoordination is observed in the histogram depicted in Fig. 14a. Many backup relays are faster than primary relays (negative separation times). The reason that explains this severe miscoordination is the selection by the authors of only one fault location at the midpoint of each line (F8 – F15, in Fig. 13). The definition of close-in fault locations to setup the LP problem is more appropriate to avoid loss of selectivity.

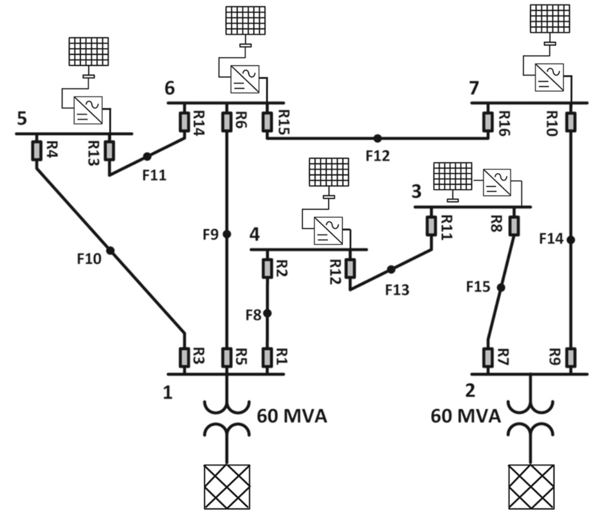


Fig. 13 Case Study 3: IEEE14 bus test system with 5 IBR DGs [18]

We corrected the reported time dial settings in [18] by running the optimization model that includes transient states [8, 16]. The improved coordination solution for 30 uniformly distributed fault locations per line is $\mathbf{D}^{*}_{\mathbf{d}}$ = [0.3383; 0.3531; 0.3504; 0.4170; 0.1762; 0.4549; 0.3682; 0.2598; 0.4423; 0.2336; 0.3986; 0.1729; 0.2405; 0.2874; 0.2225; 0.5803]. With these new settings, selectivity is improved from 79.43 % to 99.8 %, and as we can see in Fig. 14b all miscoordination and non-selective occurrences observed in Fig. 14b no longer exist. It is worth noting that the standard deviations of the distributions depicted in Fig. 14 are also improved from 24.39s to 2.39s.

This means that optimal settings with transient states $\mathbf{D}^{*}_{\mathbf{d}}$ produce a distribution of separation times Eq.(3) with a lower dispersion around an average value of 0.77s. This result highlights the importance of the inclusion of transient states in the optimization model in order to avoid miscoordination that may jeopardize power system security.

4.4 Summary of results and final remarks

In Table 2, we summarize the results of the three study cases analyzed in this paper. In all cases, the selectivity index is improved when optimization models account transient states in their formulation. As expected, speed is slightly deteriorated when we seek higher selectivity levels [31], except in case study 3.

From case study 2, we observe that when transient states are considered, it is possible to neglect IBR's lower short-circuit contributions to the interconnected system in order to achieve selectivity levels higher than 99.9 %. When traditional optimization models (without transient states) are used, the inclusion of IBR's fault contributions is mandatory

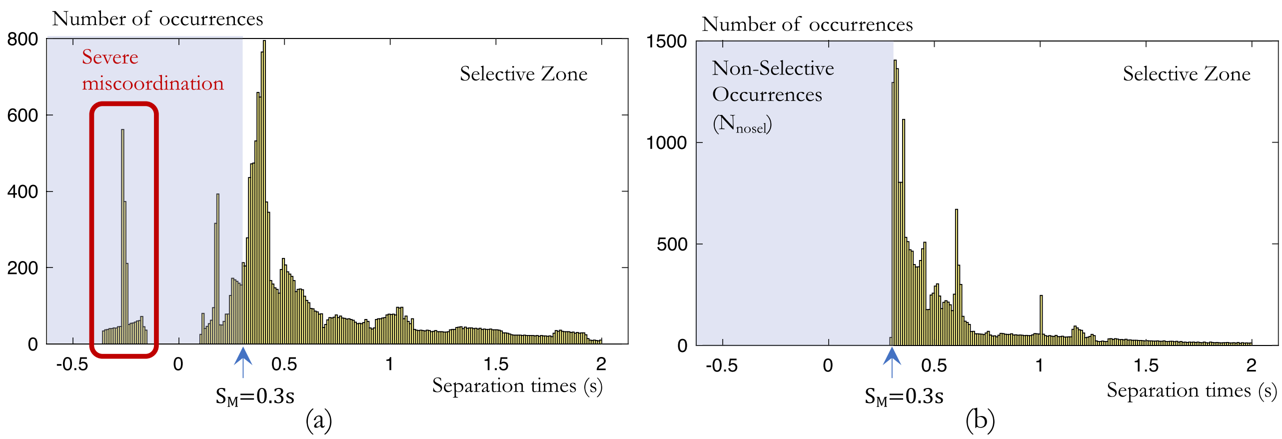


Fig. 14 Selectivity histograms for $\mathbf{D}^*$ and $\mathbf{D}_{\mathbf{d}}^*$ settings (Case Study 3)

Table 2 Summary of results

Case Study	Considers IBR?	n_f faults per line	Optimization model: Basic [1]	Optimization model: Dynamic [8, 16]	Performance indexes (Simulation) [9]: Selectivity (%)	Speed (1/s)	Sensitivity (%)
3-bus	no	2^a	*		93.2	2.40	95.15
3-bus	no	2^b		*	100	2.35	95.15
4-bus	no	2^a	*		57.0	12.00	92.60
4-bus	no	21		*	99.6	9.69	92.60
4-bus	yes	2^a	*		73.7	12.78	92.60
4-bus	yes	21		*	99.7	10.24	92.60
IEEE14	yes	1^c	*		79.4	0.88	95.40
IEEE14	yes	30		*	99.8	1.30	95.40

^{a} One closed-in fault location per relay ^{b} Two (near- and far-end) fault locations per relay in each line
^{c} One midpoint fault location per relay in each line

to improve selectivity, but is hard to achieve good results beyond 80 %.

From case study 3, we show the application of the proposed method considering a more complex system with several IBRs, whose DOCP coordination results were previously reported in the literature. In this case, as the reported solution was obtained from midpoint fault locations per line, severe miscoordination and slower speed are observed. The proposal allows to improve both selectivity and speed indexes as well.

Regarding the computational efficiency of the optimization method, we note that achieving an optimal coordination solution in [8, 16] with an acceptable selectivity level requires roughly 20–30 fault locations per line. Solving a linear programming problem, for instance, for 100 lines and 200 relay pairs (4000–6000 LP problem constraints), is today very straightforward with open-source and commercial software packages. On the other hand, the simulator required to validate the solutions must consider thousands of fault locations per line. Using a single CPU core can result in lengthy simulation time. However, this is not necessary since the simulator can be run in several cores simultaneously, as shown in [32], with a substantial reduction in CPU times.

This exercise was carried out with some limitations that simplify the description of the problem in order to make it understandable and finally demonstrate the importance of using more detailed optimization models. We assume that three-phase solid short-circuit currents are invariant values, disregarding the effects of system dynamics and IBR's specific control schemes. However, future research should encompass other important aspects such as the system dynamics, n-1 contingencies, as well as the consideration of other fault types and protection schemes (e.g., 67N, islanding modes). The development of more advanced and realistic simulators will bring the optimized settings closer to the desired performance of the DOCP system in the real world.

5 Conclusion

The severe consequences of the omission of the transient configurations in the problem of optimal coordination of directional overcurrent protections (DOCP) in interconnected power systems with inverter-based generation resources (IBR) are analyzed.

Although it has been already pointed out in some publications, it is shown that the vast majority of the published articles in the subject fail to consider this fundamental group of constraints and it is also shown that application of simplified approaches that disregard the dynamic network configuration states yields to a loss of selectivity and, therefore, to a miscoordinated operation of the protection system.

When transient states are duly included in the model, the small fault current contribution of the inverters connected to an interconnected power system can be neglected in order to achieve a DOCP coordination solution that assures a high selectivity index.

The consideration of the transient states in DOCP optimization models is imperative, and the selectivity of the protective system is significantly improved when this effect is duly included in the model.

The use of advanced, diverse optimization methods without considering these constraints is a critical aspect and must be improved and enhanced using a complete reformulation of the optimization problem that incorporates the dynamic network configuration states during the fault clearing process in order to avoid miscoordinated relay operation.

Author Contributions P.D. and A. U. conceptualize the idea. P.D. wrote the main manuscript text. All authors reviewed the manuscript.

Availability of data and materials No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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